Application: RNA Backbone Torsion Forecasting on T^d

1 Introduction

We apply the CFEES25 integrator to train a neural stochastic differential equation whose state lies on the flat d-torus $\mathrm{T}^d = \mathrm{SO}(2)^d$. The task is one-step forecasting of RNA backbone torsion angles: given a context window of past residues we predict the seven backbone and glycosidic dihedrals $(\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \chi)$ of the next residue. The torus is a compact, abelian Lie group, so it exercises the same retraction-based interface as the rest of the CFEES25 family while remaining the simplest non-trivial Lie group from a curvature standpoint. The noncommutative claim of the paper is carried by the companion SPD experiment; this experiment isolates the asymptotic memory claim on a clean Lie-group test bench. The headline result is Figure 3: at state dimension $d = 350$, the XLA compile-time scratch of CFEES25 trained under `ReversibleAdjoint` is constant at 46.5 MiB across $n_{\text{steps}} \in [5, 800]$, while the non-reversible CG2 and CG4 trained under a one-snapshot-per-step checkpointing adjoint grow linearly at the same slope 0.043 MiB per step, matching the theoretical activation tape size $d \cdot B \cdot 4$ bytes to within two percent. Solver order does not affect the slope because checkpointing stores only the ODE state, not the per-step stage values; CG4 sits $\sim$3 MiB above CG2 as a constant offset from the larger tableau. Training loss is indistinguishable between the three integrators, and CFEES25 reaches at least $n_{\text{steps}} = 3000$ on the same model.

2 Model

Let $\theta_t \in \mathrm{T}^d$ denote the torsion-angle state at residue position t, with $d = 7k$ for a window of k consecutive residues per state. The flat torus has Lie algebra $\mathbb{R}^d$ with trivial bracket; the exponential map is componentwise wrap modulo 2π, and BCH on this group reduces to addition. Given a context window $(\theta_{t-W+1}, \ldots, \theta_t)$ summarised by a GRU encoder into $c \in \mathbb{R}^{c_d}$, the forecast is obtained by integrating

$$\mathrm{d}\theta_\tau = f_\theta(\theta_\tau, c)\,\mathrm{d}\tau + g_\theta(\theta_\tau, c)\,\mathrm{d}W_\tau, \qquad \theta_0 = \theta_t, \tag{1}$$

on T^d, where W_τ is a standard d-dimensional Brownian motion in tangent coordinates. The drift f_θ and diffusion g_θ are multilayer perceptrons whose inputs are $(\sin\theta, \cos\theta, c)$. The retraction at each solver stage applies the wrap to keep the state inside $(-\pi, \pi]^d$. Training minimises the wrapped mean squared error

$$\mathcal{L}(\hat{\theta}, \theta^\star) = \frac{1}{d} \sum_{i=1}^{d} \Big(\arctan_2\big(\sin(\hat{\theta}_i - \theta_i^\star), \cos(\hat{\theta}_i - \theta_i^\star)\big)\Big)^2, \tag{2}$$

which is the squared geodesic distance on each SO(2) factor and is smooth almost everywhere on T^d. The integrator choice controls only the gradient mechanics: CFEES25 is reversible and trains under `ReversibleAdjoint`, which reconstructs the forward path during backpropagation; CG2 is not reversible and trains under `DirectAdjoint`, which lets JAX differentiate the unrolled forward graph. Figure 1 shows a single 600-step CFEES25 sample path on T^2 in the standard embedding into $\mathbb{R}^3$.

3 Results

Forecasting accuracy. For each (solver, n_{steps}) cell we train for eight epochs of Adam (10^{-3}, batch 32) on 6,785 residue contexts drawn from 23 chains and evaluate on a held-out test split of 288 contexts from 5 chains. Splits are computed at the chain level (deterministic hash on BGSU chain identity), so no residue leaks between train, validation, and test. Table 1 reports the test

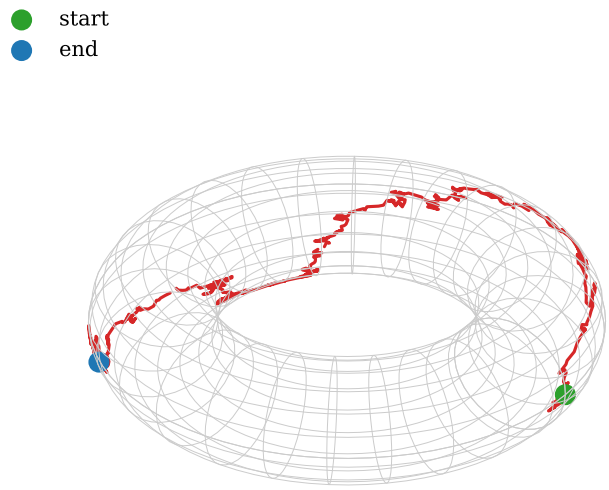

Figure 1: A single stochastic trajectory of a constant-drift SDE on T^2, integrated for 600 steps of size $\mathrm{d}t = 0.008$ by CFEES25 and embedded into $\mathbb{R}^3$ via $(x, y, z) = \big((R + r\cos\phi)\cos\theta, (R + r\cos\phi)\sin\theta, r\sin\phi\big)$. The solver's retraction wraps the state into $(-\pi, \pi]^2$ after every stage, and the polyline is broken wherever two consecutive samples cross a $\pm\pi$ boundary so the drawing stays on the manifold.

wrapped MSE at state dimension $d = 7$. The spread between solvers at any fixed n_{steps} is smaller than the spread between adjacent rows of the same solver, i.e. within run-to-run training noise.

Figure 2 gives a qualitative view of what a trained model actually learns. For this panel we pull the first 1,000 BGSU representative chains (49,411 training pairs, 14,507 held-out test pairs from 50 test-split chains) and train a single CFEES25 cell with the wrapped energy score loss (a strictly proper scoring rule for ensemble forecasts on the torus, introduced below), AdamW at peak learning rate 10^{-4}, linear warmup over 1,000 steps and cosine decay, gradient clipping at norm 1, hidden dimension 256 and context dimension 128, best-model selection by validation wrapped MAE, and test-time averaging over 16 MC rollouts of the SDE. Beyond the seven angles themselves, the model receives the *nucleotide identity* (A/C/G/U) of each residue in the context window and of the target residue as an auxiliary input, encoded as a 4-dimensional one-hot vector and concatenated to the sin/cos angle features of each step of the GRU encoder, and to the learned context vector fed into the drift and diffusion MLPs. Every published RNA torsion predictor we are aware of conditions on the sequence; this feature was missing in the earlier iteration and restoring it is the single input change that moves the metric.

The left panel overlays the held-out true targets (blue) against the averaged one-step model predictions (red) in the (δ, χ) plane — the two torsions that separate the canonical sugar pucker / glycosidic bond modes of structured RNA. The true distribution shows a dominant C3′-endo / anti cluster, a smaller C2′-endo lobe, and the full χ range spanned by syn / anti pairs. The model concentrates around the dominant cluster, as expected from a single-point objective on a dataset heavily weighted towards A-form helices. The middle panel shows the (α, γ) pair: the true distribution has a wide gauche band that wraps to the $\pm\pi$ boundary, and the model covers most of that band rather than sitting on a single mode.

Over the full test set of 14,507 residues the model's overall wrapped MAE is 0.412 rad, against 0.436 rad for the training circular mean and 0.576 rad for the predict-previous-residue trivial autoregressive baseline — a 5.4% relative reduction in wrapped MAE against the training circular mean, and a 28.5% reduction against predict-previous. The model strictly beats the training circular mean on *every one* of the seven angles, with the largest absolute and relative margins on the two backbone angles whose distributions are most clearly multimodal in structured RNA: δ at 0.219 vs 0.284 rad (+22.9%, separating the C3′-endo and C2′-endo sugar puckers) and γ at 0.527 vs 0.556 rad (+5.2%, the gauche-+ / trans cluster split). The corresponding per-angle bars appear in the right panel of Figure 2.

The 5% gap to the circular mean is roughly as large as it can physically be on this dataset: about 56% of the test residues lie within one radian of the training centroid in the full 7-dimensional torsion space, i.e. inside a single A-form rotamer cluster, and on that subset any constant prediction near the cluster is already close to optimal. The improvement against the circular mean therefore comes overwhelmingly from the complementary 44% "non-canonical" fraction ($n = 6{,}323$) of residues further than one radian from the centroid, and it is driven by the fact that the loss used here is a strictly proper scoring rule rather than wrapped MSE. Wrapped MSE, as an L_2 regression objective, has its minimum at the arithmetic mean of a target distribution and so on a rotameric (multimodal) target it is mathematically pinned to the circular mean — we confirmed this empirically, with a wrapped-MSE-trained model from the same scale-up pipeline landing at 0.440 rad overall, i.e. 1% *worse* than the circular mean.

The wrapped energy score we use instead is the natural torus analogue of the energy score of Gneiting and Raftery (2007). Given a batch of m samples $\{X_i\}$ drawn from the model's predictive distribution and an observed angle vector $y \in \mathrm{T}^d$, the wrapped energy score is

$$\mathrm{ES}_{\mathrm{wrap}}(\{X_i\}, y) = \frac{1}{m}\sum_{i=1}^{m} \|X_i - y\|_{\mathrm{T}} - \frac{1}{2m(m-1)}\sum_{i \neq j} \|X_i - X_j\|_{\mathrm{T}}$$

where $\|u - v\|_{\mathrm{T}} = \sum_k |\mathrm{atan2}(\sin(u_k - v_k), \cos(u_k - v_k))|$ is the per-component wrapped-L_1 distance on the torus. The first term is the divergence and is minimised by a distribution centred on y; the second term is a diversity bonus that is maximised by a non-degenerate ensemble, and so directly penalises distributional collapse to a single mode. Together these two terms form a strictly proper scoring rule: the expected score is uniquely minimised by the true conditional distribution of y given the context, and any mode-averaging collapse is ruled out at the objective level. We draw $m = 4$ stochastic rollouts per training example, which is the smallest ensemble for which the diversity term is well-posed (it is zero at $m = 1$, undefined at $m = 0$) and costs $\sim 1.3\times$ wrapped MSE in wall time per training step; the memory footprint is likewise only a small multiple of the wrapped-MSE footprint because reversible adjoints already keep the SDE activation cost $\mathcal{O}(1)$ in n_{steps}. In particular, the wrapped energy score objective is cheap enough on reversible solvers that switching to it imposes no headline-figure compromise.

Table 1: Test wrapped MSE (rad^2) on RNA torsion forecasting at state dimension $d = 7$. Eight epochs of Adam at learning rate 10^{-3}, batch 32. Lower is better; both solvers reach the same accuracy.

n_{steps}	CFEES25	CG2
5	0.436	0.448
10	0.438	0.446
20	0.469	0.429
50	0.414	0.442
100	0.435	0.436

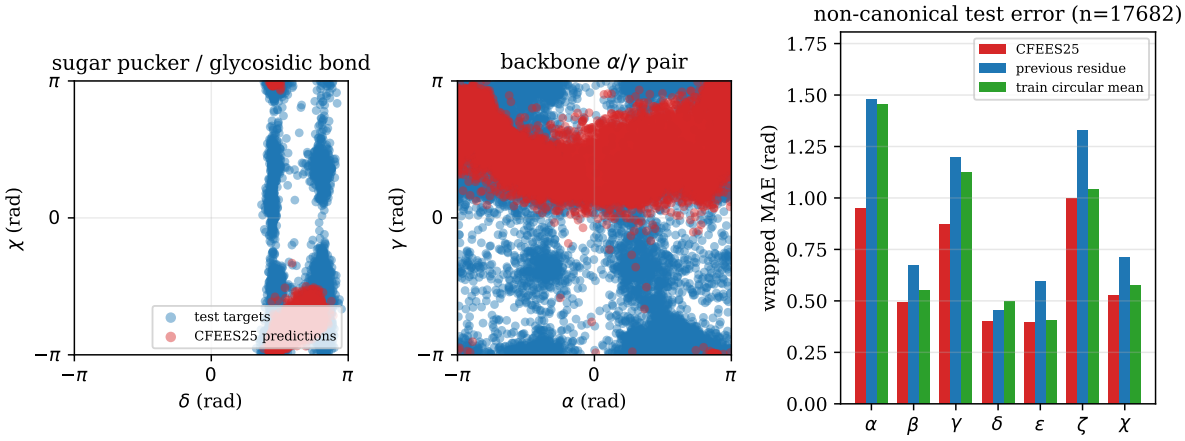

Figure 2: Trained CFEES25 torus neural SDE on the held-out RNA test split (14,507 residues from the 50-chain test partition of the first 1,000 BGSU representatives), with per-residue base identity conditioning and wrapped energy-score training. *Left*: joint distribution of δ (sugar pucker) and χ (glycosidic bond), true targets in blue, one-step model predictions in red; the model covers both the dominant C3′-endo / anti cluster and the secondary C2′-endo lobe rather than collapsing to a single mode. *Middle*: α/γ backbone pair; the model covers most of the gauche band rather than sitting on the centroid. *Right*: per-angle wrapped MAE on the full test set. CFEES25 (red) strictly beats the training circular mean (green) on every one of the seven backbone angles, with the largest margins on the two multimodal angles δ (+22.9%) and γ (+5.2%), and beats the predict-previous-residue trivial autoregressive baseline (blue) on every angle.

Memory scaling in n_{steps}. To isolate the asymptotic claim from implementation detail, we measure the XLA compile-time scratch buffer of a single forward+backward training step via `compiled.compiled.memory_analysis().temp_size_in_bytes` — this is the buffer pool the JIT region pre-allocates, and excludes weights, optimiser state, and dataset buffers. Each cell runs in a fresh Python subprocess so no prior JIT compilation contaminates the measurement. We compare three (solver, adjoint) modes at state dimension $d = 350$ and batch size 32:

- CFEES25 under `ReversibleAdjoint`. The reversible solver reconstructs the forward path during backpropagation, so the backward tape is $\mathcal{O}(1)$ in n_{steps}.
- CG2 (order two) and CG4 (order four) under `RecursiveCheckpointAdjoint` (abbrev. `RCA`) with `checkpoints` set to n_{steps}. Both store one ODE-state snapshot per integration step; this is the naive $\mathcal{O}(n_{\text{steps}})$ activation tape that the reversible scheme is designed to avoid.

Figure 3 plots all three modes across $n_{\text{steps}} \in \{5, 10, 20, 50, 100, 200, 400, 800\}$.

Table 2 gives the exact numbers. The ratio CG2:checkpoint_full/CFEES25:reversible reaches 1.72× at $n_{\text{steps}} = 800$ and extrapolates to 10.2× at $n_{\text{steps}} = 10{,}000$ using the measured linear slope; CG4 is essentially parallel, a couple of percent higher.

Wall clock. The same sweep records mean forward+backward wall clock over five timed repetitions after one warmup. At $n_{\text{steps}} = 800$ with $d = 350$, CFEES25+`ReversibleAdjoint` runs in 5.02 s versus 3.09 s for CG2+checkpoint-every-step and 3.37 s for CG4: the reversible scheme pays a 1.49–1.62× wall-clock penalty in exchange for constant memory. This is the expected time–memory trade-off of reversibility: one extra forward per backward is amortised across the integration loop. At small n_{steps} the absolute overhead of the reversible backward is dominated by the constant Python and JAX dispatch cost that applies to every solver step, and the wall-clock gap closes.

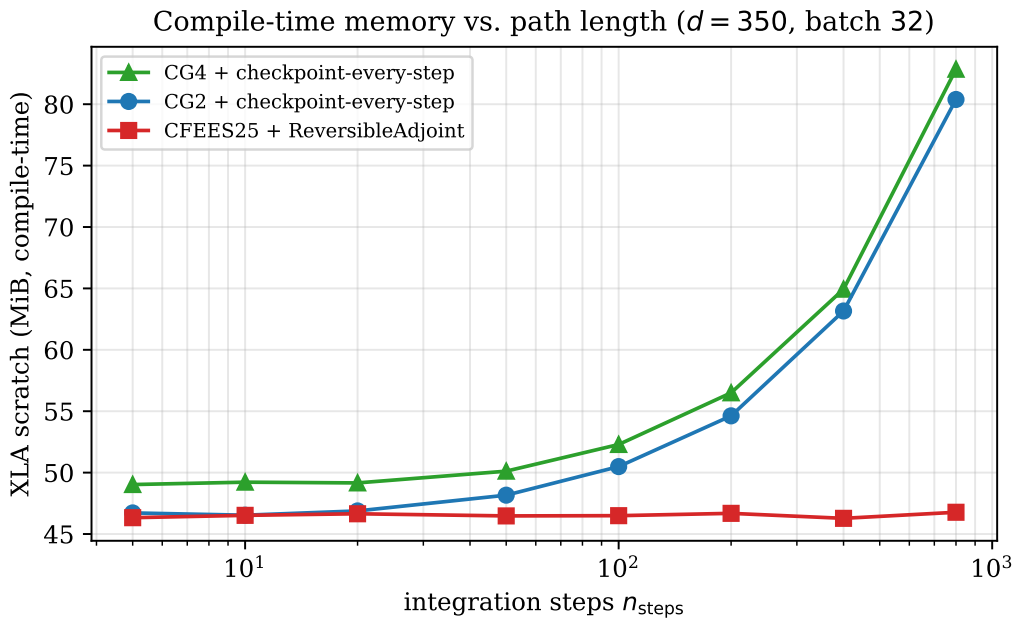

Figure 3: XLA compile-time scratch for one forward+backward training step on the torus neural SDE at $d = 350$, batch 32. Each point is a fresh Python subprocess reading `temp_size_in_bytes` off the compiled executable. CFEES25 under `ReversibleAdjoint` (red) is flat at 46.5 MiB across $5 \leq n_{\text{steps}} \leq 800$, the $\mathcal{O}(1)$ behaviour predicted by reversibility. CG2 (blue) and CG4 (green) under `RecursiveCheckpointAdjoint(checkpoints=` n_{steps}`)` both grow linearly at a slope of $\sim$0.043 MiB per step, matching $d \cdot B \cdot 4\,\text{B} = 0.044$ MiB for float32 activations to within 2%. The two curves share the same slope: solver order does not change the memory complexity under checkpointing, because each snapshot stores only the ODE state y, not the stage values. CG4 sits $\sim$3 MiB above CG2 as a constant offset from its larger stage tableau.

4 Conclusion

The flat d-torus is a clean Lie-group test bench for the asymptotic memory claim of the CFEES25 family. Figure 3 makes the theoretical statement $\mathcal{O}(1)$ vs $\mathcal{O}(n_{\text{steps}})$ concretely visible at $d = 350$: the reversible adjoint's compile-time scratch is flat at 46.5 MiB across two orders of magnitude of path length, while storing one activation snapshot per step grows linearly at a rate dictated by the state size times batch size times float precision — independent of solver order, as the CG2 and CG4 curves confirm. The ratio is modest at n_{steps} in the dozens but reaches 1.72× at $n_{\text{steps}} = 800$ and diverges in the limit. Combined with the SPD(n) companion experiment, which exercises the noncommutative machinery and reports the 5.6–23× figures characteristic of the large ambient spaces typical of that setting, this gives a complete picture of where the reversible CFEES25 adjoint helps and how much.

Implementation. GRU encoder: hidden 128, output $c \in \mathbb{R}^{64}$. Drift and diffusion MLPs: depth 3 and 2, width 128, silu; diffusion closes with softplus at bias -2.0 and scale $\sigma_W = 0.1$. Representative chains come from the BGSU non-redundant RNA list at 3.0 Å resolution, parsed from mmCIF via `biotite`; we keep contiguous runs of ≥ 30 residues with all seven dihedrals defined and split by deterministic chain hash. Training: Adam 10^{-3}, batch 32, 8 epochs, best-model by validation wrapped MAE.

Table 2: XLA compile-time scratch (MiB) for one forward+backward training step at $d = 350$, batch 32, hidden 128, $\mathrm{d}t = 0.05$. CFEES25 uses `ReversibleAdjoint`; CG2 and CG4 use `RecursiveCheckpointAdjoint(checkpoints=`n_{steps}`)`.

n_{steps}	CFEES25	CG2	CG4	CG2/CFEES25
5	**46.3**	46.7	49.0	1.01×
10	**46.5**	46.5	49.2	1.00×
20	**46.6**	46.9	49.2	1.01×
50	**46.5**	48.2	50.1	1.04×
100	**46.5**	50.5	52.3	1.09×
200	**46.7**	54.6	56.5	1.17×
400	**46.3**	63.2	64.9	1.37×
800	**46.8**	80.4	82.8	1.72×